

Electron Thruster for Small Spacecraft

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Abstract

Atmospheric drag can limit the orbital lifetime of small spacecraft, while their size and power budgets constrain propulsion hardware. This paper presents an electron propulsion concept that uses the ionosphere to supply its return current, eliminating the need for stored propellant. A cathode biased below the spacecraft body emits an electron beam through an aperture; the body charges positively and collects ambient electrons to replenish the emitted charge. Particle-in-cell simulations (WarpX) of conducting bodies 10 mm in diameter predict 3.4-43 nN of beam thrust at the tested supply voltages of 100-350 V, with beam-supply powers of 12-278 mW and escape fractions of 96-99 %. A model calibrated to these simulations and coupled to year-long orbital and ionospheric calculations estimates the power required for drag compensation. For a body 10 mm in diameter and 30 mm long in sun-synchronous orbits at 500-600 km, the estimated mean beam-supply demand is 10-46 mW near solar maximum and 0.4-2 mW near solar minimum. The architecture combines electron emission with ambient current collection in the spacecraft's own structure, offering a compact propulsion approach for missions with nanonewton thrust requirements.

1 Introduction

Conventional gridded ion thrusters require stored propellant, a feed system and a neutralizer [1]. This hardware adds mass, volume and integration complexity. This paper investigates a simpler architecture: emit electrons and let the ionosphere provide the return current.

The tradeoff is thrust per watt. For a directed exhaust, $F/P = 2\eta/v_e$, where η is the electrical-to-jet efficiency and v_e is the exhaust speed. Electrons accelerated through a few hundred volts reach high exhaust speeds, giving little momentum per unit energy. The simulated 200 V operating point delivers about 0.2 $\mu\text{N/W}$, making electron propulsion power-intensive at millinewton thrust levels.

At the nanonewton drag levels examined here, the beam-supply demand is milliwatts. A cathode, an aperture, a high-voltage supply and the spacecraft's conducting skin provide propulsion without stored propellant. The aim is to reduce the mass, volume and integration complexity of station-keeping hardware on small spacecraft.

2 Concept

A cathode held at $-V$ with respect to the spacecraft body emits electrons and accelerates them out through an aperture (Fig. 1). The escaping beam carries directed momentum and produces reaction thrust. Each escaping electron leaves the body one charge more positive, so the body's conducting skin collects ambient electrons from the plasma, and the supply returns them to the cathode.

The body approaches the floating potential φ , measured relative to the ambient plasma, where net collection balances beam escape.

The current loop relies on an energy asymmetry: accelerated beam electrons can escape the positive body's potential well, while low-energy ambient electrons are attracted to it. Ambient electrons arrive over the collecting surface rather than as a directed counter-beam. Thus the same spacecraft structure acts as both the accelerating electrode and the return-current collector.

3 Theory

The propulsion mechanism couples the momentum carried by the emitted beam to the current collected from the surrounding plasma.

Thrust is the momentum flux of the escaping beam:

$$F = \frac{I_{\text{esc}} \sqrt{2m_e \text{KE}}}{e}. \quad (1)$$

Here $I_{\text{esc}} = f_{\text{esc}} I$ is the escaping part of the emitted current; collimated exhaust is assumed.

The energy each electron actually leaves with is

$$\text{KE} = \kappa e (V - \varphi), \quad (2)$$

with KE in joules and V the supply voltage.

Jet power divided by electrical beam-supply power is

$$\eta = \frac{P_{\text{jet}}}{P} = \kappa \frac{V - \varphi}{V} f_{\text{esc}}. \quad (3)$$

Throughout, $P = VI$ denotes power delivered by the supply between body and cathode. Total spacecraft power also depends on emitter technology and converter efficiency.

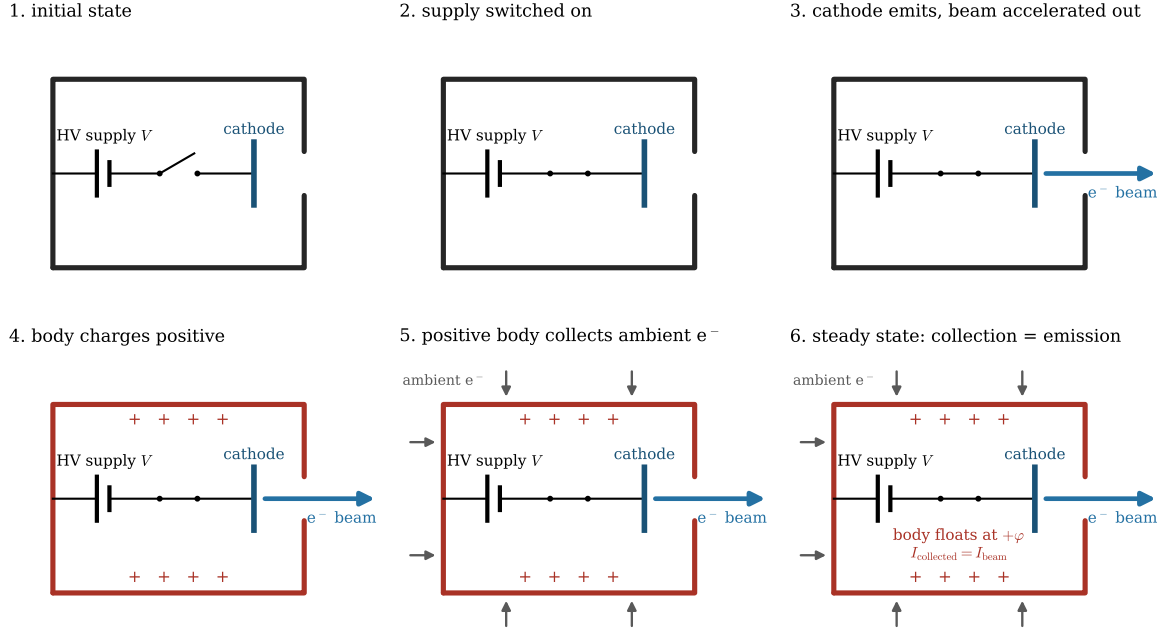


Figure 1: Operating sequence. The supply is switched on, the cathode emits, the escaping beam charges the body positive, and the positive body collects ambient electrons until collection equals emission.

The energy fraction κ relates exhaust energy to the available potential drop; ideal emission at cathode potential gives $\kappa \approx 1$. The floating potential φ depends on collecting area, body shape and plasma conditions. The ratio φ/V is the fraction of supply voltage spent overcoming the body's positive potential. The escape fraction f_{esc} is the beam fraction that clears the body, controlled by the exit-aperture geometry.

Control. The electrical operating point provides two useful control signals:

1. **Thrust estimation.** Measured current, supply voltage and body potential relative to the plasma give a thrust estimate through Eqs. 1–2, with calibrated escape and divergence corrections. An in-flight thrust stand is unnecessary.
2. **Feedback.** A microcontroller can adjust the supply voltage to track the thrust command using the measured current and floating potential. Density, temperature and day/night changes enter through these signals, without an explicit ionosphere model or lookup table.

This is a proposed control principle; flight implementation requires current and plasma-potential sensing and calibration of the thrust estimate.

4 Particle-in-cell simulations

The device is simulated with WarpX [2, 3] in axisymmetric (RZ) electrostatic mode. The slender body is the main case for the simulation results and station-keeping analysis; the compact body provides the verification-ladder reference and a geometry comparison. Both demonstrate

Table 1: Parameters of the RZ simulations. Voltage and current are sampled operating points, not limits of the propulsion concept.

Parameter	Value
Body diameter	10 mm
Body length, slender; compact	30.5 mm; 5.5 mm
Wall, lid, floor thickness	0.4 mm
Cathode; emitting-spot radius	1.5 mm; 0.5 mm
Aperture radius; gun gap	2.0 mm; 4.7 mm
Launch offset above cathode	0.30 mm (two cells)
Supply voltages sampled	100, 200, 300, 350 V
Prescribed beam current	0.121–0.793 mA
Ambient electron density	$1.627 \times 10^{12} \text{ m}^{-3}$
Electron; ion temperature	1319 K; 936 K
Reduced ion mass	$400 m_e$
Cell size	$\Delta r = \Delta z = 0.15 \text{ mm}$
Grid, slender; compact	200×608 ; 200×440
Ambient macroparticles per cell and species	16
Supply on; beam on	100 ns; 150 ns
Run duration	800 ns

the same propulsion mechanism. Figure 2 shows their shared internal gun geometry, and Table 1 gives the simulation parameters.

The acceleration gap is 4.7 mm in both configurations. Extending the spacecraft body from 5.5 to 30.5 mm increases the return-current collecting area without enlarging the acceleration region or frontal area. The compact body is the smallest complete electrode-and-collector geometry tested here; a minimum size has not been established. These dimensions characterize the electrodes and collecting structure; supply and emitter packaging depend on the flight implementation.

The body potential evolves self-consistently from the net charge emitted and collected, divided by the body's

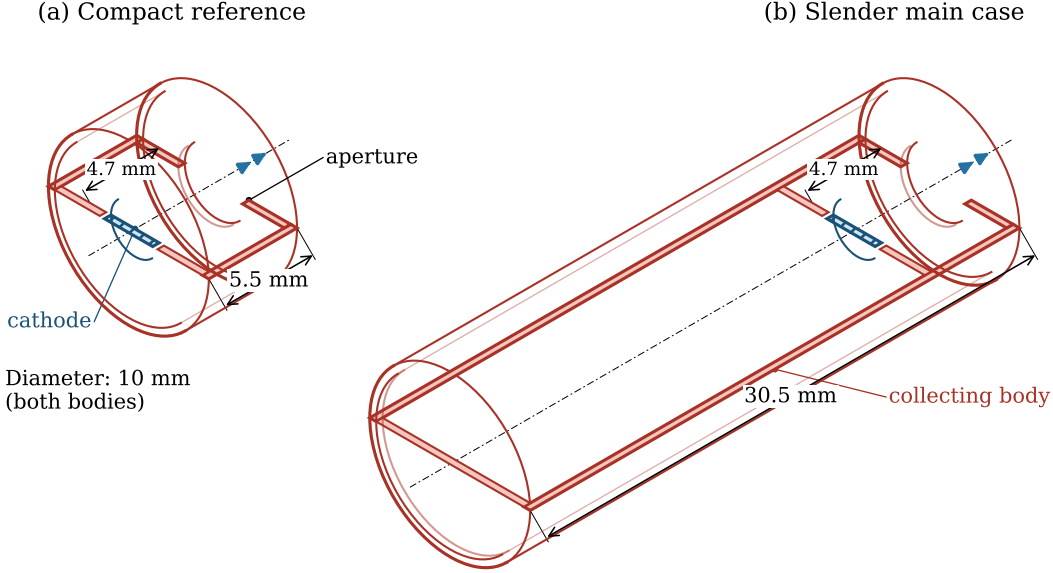


Figure 2: Simulated conducting bodies at the same scale: (a) the compact reference, 5.5 mm long; (b) the slender main case, 30.5 mm long, used for the station-keeping calibration. Both are 10 mm in diameter, with a 4 mm aperture, a 3 mm cathode disk and a 4.7 mm cathode-to-lid gap. The additional length provides collecting surface around the same acceleration region. Red denotes the collecting body and blue the cathode and escaping beam. The supply holds the cathode at $-V$ relative to the body.

self-capacitance. Open boundaries and ambient-particle replenishment represent the surrounding ionosphere. A nine-stage verification sequence builds from a vacuum electron gun to the full device, checking emission, collection and floating-body coupling against theory and fixed criteria on escape, current balance and momentum; thermal collection agrees with theory within 1 %.

Figure 3 shows the slender-body simulation at 200 V: the beam density at the final time, beam fates, thrust and beam-supply power. Table 2 summarizes the sampled operating points. The fitted thrust relation is $F = 3.27 I \sqrt{KE}$ (with F in nN, emitted I in mA and KE in eV; the slope includes interception and divergence), with an energy calibration $\kappa \approx 0.81$. Increasing the body length multiplies its collecting area by 3.5: at 200 V, the floating potential falls from 17.0 to 4.4 V and thrust rises from 13.65 to 14.22 nN. Collected-particle momentum offsets 0.4-1.4 % of beam thrust in the 200-350 V cases. A separate 3D comparison with the geomagnetic field across the beam at flight strength changes thrust by under 1 %.

The fitted κ reflects the numerical launch position: electrons are introduced 0.3 mm above the cathode, missing part of the accelerating potential drop. A vacuum field calculation places this plane about 19 % of V above cathode potential; the simulated exhaust-energy deficit is 17-18 % of V in the cases of Table 2. Emission at cathode potential is therefore expected to increase κ toward the ideal limit of one. Direct PIC confirmation is still required. All reported performance and mission estimates retain the existing calibration.

Table 2: Simulated operating points for the slender main case and compact reference. $\eta = KE f_{\text{esc}} / (eV)$. Grid sensitivity: $F \pm 4 \%$, $\varphi \pm 2 \%$.

body	V V	I mA	φ V	F nN	P mW	f_{esc} %	η
slender	200	0.342	4.4	14.22	68.4	98.4	0.79
10×30.5 mm	350	0.793	14.0	43.33	278	99.1	0.77
compact	100	0.121	5.4	3.42	12.1	96.1	0.74
10×5.5 mm	200	0.342	17.0	13.65	68.4	98.4	0.73
	300	0.630	36.3	30.13	189	99.0	0.69
	350	0.793	48.3	40.48	278	99.1	0.68

5 Station keeping

The station-keeping calculations use the slender body, with its collection prefactor fitted to the 200 and 350 V slender PIC cases. The mission geometry approximates the 30.5 mm simulated body as a 10 mm diameter, 30 mm long cylinder of mass 3.1 g, flown with its axis along the velocity so that the thrust axis and low-drag pose coincide.

Drag and the plasma along the orbit come from a year-long TudatPy propagation [4] with NRLMSISE-00 density [5] driven by measured solar and geomagnetic indices and IRI-2020 plasma [6] along the orbit, near solar maximum (2024) and minimum (2019). With the thruster on, ideal drag cancellation defines the thrust demand at each 5-minute point; the supply voltage follows it at the least beam power, with the floating potential estimated from the local plasma using the simulation-calibrated collection model. With the thruster off, the orbit decays freely. These estimates extend the calibration to voltages and plasma conditions beyond the PIC samples; 350 V marks the highest tested voltage, not a physical limit of

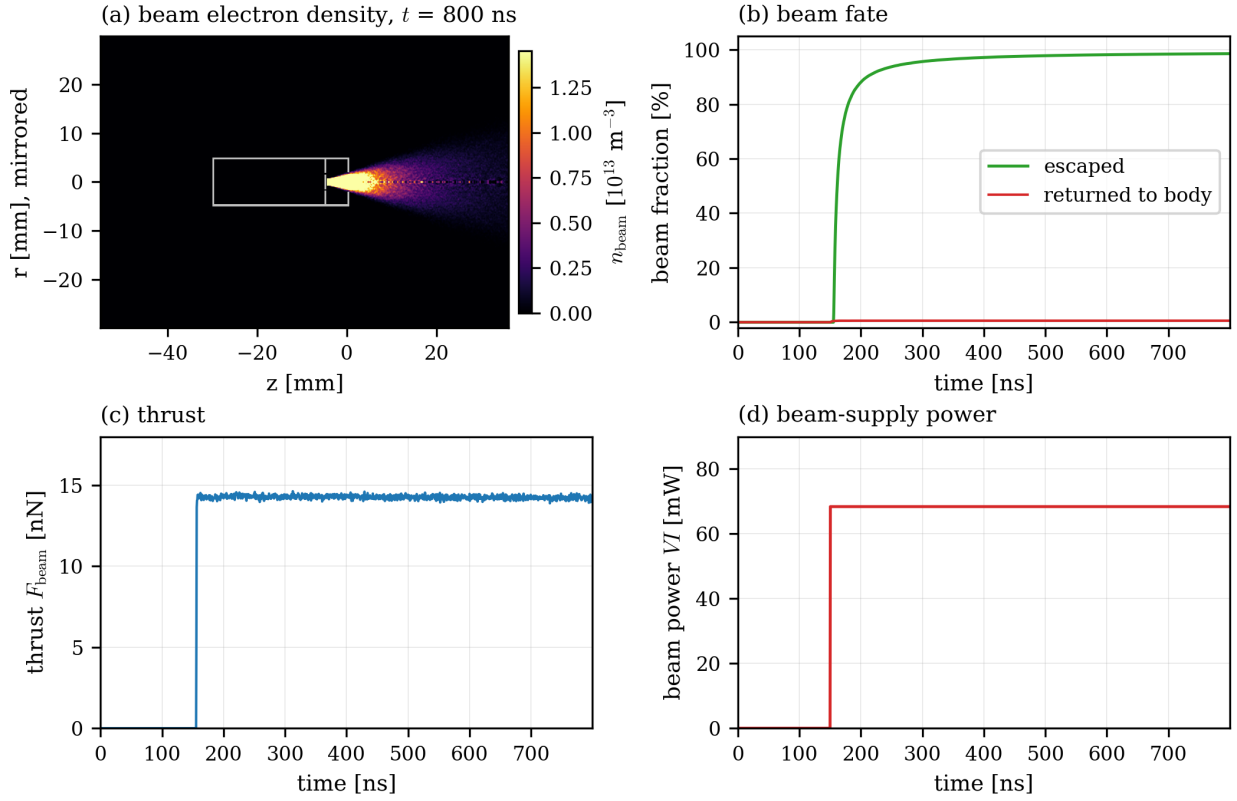


Figure 3: PIC results for the 200 V slender-body case, using the run shown in the repository animation. (a) Beam electron density at 800 ns, mirrored about the symmetry axis. (b) Cumulative fractions escaped and returned to the spacecraft; the remainder is still in the domain. (c) Escaping axial momentum flux, giving approximately 14.2 nN of reaction thrust. (d) Beam-supply power $VI = 68.4$ mW during prescribed emission at 0.342 mA. The density is a final snapshot; the other panels show the run history.

Table 3: Modeled orbital decay and station keeping on a sun-synchronous orbit (10:30 ascending node), for a year near solar maximum (2024, dense upper atmosphere) and a year near solar minimum (2019, thin upper atmosphere). Lifetime is how long the orbit takes to decay from the listed altitude to 120 km with the thruster off; > 5 yr means it had not decayed that far within the five-year propagation. Drag is the year-mean drag force, which is the thrust the thruster must supply to hold the listed altitude; power is the year-mean beam-supply power that delivers it. The 1U drag follows from the assumed drag-area ratio ($\times 103$); its power is estimated using the same scaling. [†]47 % of the points require extrapolation above the highest simulated supply voltage, 350 V.

altitude	3 g slender can (10 × 30 mm)			1U CubeSat (1.33 kg), estimate		
	lifetime, thruster off	drag [nN]	power to hold [mW]	lifetime, thruster off	drag [μ N]	power to hold [W]
<i>2024, near solar maximum</i>						
400 km	46 d	38	278 [†]	167 d	3.9	29 [†]
500 km	204 d	8.6	46	2.1 yr	0.89	4.8
550 km	329 d	4.3	20	> 5 yr	0.45	2.1
600 km	2.2 yr	2.3	9.6	> 5 yr	0.23	0.98
650 km	> 5 yr	1.2	4.8	> 5 yr	0.12	0.49
700 km	> 5 yr	0.66	2.5	> 5 yr	0.068	0.26
<i>2019, near solar minimum</i>						
400 km	241 d	3.8	21	2.4 yr	0.39	2.2
500 km	3.5 yr	0.49	2.0	> 5 yr	0.050	0.20
550 km	4.5 yr	0.21	0.78	> 5 yr	0.021	0.080
600 km	> 5 yr	0.10	0.37	> 5 yr	0.010	0.039
650 km	> 5 yr	0.057	0.21	> 5 yr	0.006	0.022
700 km	> 5 yr	0.037	0.14	> 5 yr	0.004	0.014

the concept.

Table 3 gives the model results. Without the thruster the 3 g body re-enters from 400 km in 46 days and from 550 km in 11 months near solar maximum. At 500-

700 km, the mean thrust demand is 0.7-9 nN, with mean beam-supply demand of 2.5-46 mW near maximum and at most 2 mW near minimum. At 400 km near maximum the 38 nN mean is below the 43 nN simulated at

350 V, but the thin night-side plasma pushes nearly half the points above the tested voltage range; near minimum the mean demand is 21 mW. The ISS and equatorial orbits differ from the sun-synchronous one by about 10 %.

Over each orbit the demand changes by $1.6\text{--}2.0\times$ between the sunlit and eclipsed halves at 500-700 km, and on the geomagnetic-storm days of 11 May, 12 August, 7-8 and 10-11 October 2024 the daily mean reaches $2.2\text{--}3.5\times$ the annual mean, the larger factors at the higher altitudes. A separate altitude-hold calculation caps the drive at 350 V to test operation within the sampled voltage range. Temporary thrust deficits produce altitude dips of less than 10 m at 500-700 km over the modeled year.

For comparable exhaust energy and collection performance, the required current and beam power scale with the drag area. The modeled 1U CubeSat has 103 times the can's drag area and therefore needs 103 times its thrust. Applying the same factor to power gives a first-order estimate of about 1 W at 600 km near solar maximum and tens of milliwatts near minimum. Collection at this larger scale requires separate validation. The 1U lives longer in free fall because of its larger mass-to-drag-area ratio: 167 days from 400 km near solar maximum, against 46 for the can.

6 Conclusion

A cathode, a high-voltage supply and the spacecraft's conducting skin form an electron propulsion system that replenishes emitted charge from the ionosphere. The simulated geometries produce 3.4-43 nN at beam-supply powers of 12-278 mW. The compact-to-slender comparison shows how collecting area can increase around an unchanged acceleration region. Mission calculations for the slender body indicate station keeping for a 3 g spacecraft at 500-700 km with milliwatt mean beam-supply demand. The contribution is a compact propulsion architecture in which the spacecraft structure closes the current loop. Cathode-potential emission, long-time settling and collection at larger spacecraft scales remain the principal simulation questions for extending the quantitative results.

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